Enhancing Satellite-Based Wildfire Monitoring: Advanced Contextual Model Using Environmental and Structural Information

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***Abstract*— Satellite-based wildfire detection models provide real-time data that is essential for effective management of wildfires, whose frequency and intensity are increasing. However, conventional temperature anomaly-based contextual algorithms fail to incorporate environmental and structural information, resulting in poor generalization capabilities across various environmental conditions and wildfire intensities. This study developed a novel machine learning-based wildfire detection model incorporating advanced contextual features to fulfill the deficiencies of conventional contextual (CC) algorithms. Specifically, two advanced contextual features—considering the baseline environment (ACE) and structural similarity (ACS)—were introduced, and their effectiveness was assessed relative to CC features. The integration of ACE and ACS significantly enhanced the performance of all the evaluated machine learning algorithms, with the light gradient boosting machine (LGBM) model, considering ACE and ACS as input variables, achieving the highest accuracy (precision: 98.56%; recall: 97.78%; *F*1-score: 98.17%). Owing to its heightened detection sensitivity, the LGBM model identified additional wildfires that the GEO-KOMPSAT-2A (GK-2A) forest fire (FF) products had failed to detect. The variations in the environmental heterogeneity and wildfire intensity significantly diminished the effectiveness of conventional temperature anomaly-based algorithms. ACE and ACS effectively mitigated performance deterioration attributable to environmental heterogeneity and intensity variations, respectively. This investigation analyzed the potential error sources in conventional satellite-based wildfire detection models and applied a novel methodology to refine them, ultimately proposing a model that can be generalized across varied environmental conditions and wildfire intensities.**

***Index Terms*— Contextual features, environmental correlation, geostationary satellite, machine learning, structural similarity, wildfire detection.**

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I. Introduction

WILDFIRES are natural hazards that can occur in various types of combustible vegetation, including forests, shrubs, savannas, grasses, and croplands [1]. These fires not only destroy ecosystems by reducing vegetation cover but also cause significant economic losses by damaging forests and crops [2], [3], [4]. Moreover, the substantial emissions of aerosols and carbon dioxide from wildfires contribute significantly to air pollution and climate change [5], [6]. The frequency and intensity of wildfires are escalating annually because of global warming, resulting in increasingly diverse fire damages [7], [8], [9], [10].

Conventional field observations, which depend on site visits for data collection after a fire has been extinguished, are inadequate for proactive responses to the rapidly evolving nature of wildfires. In contrast, satellite-based wildfire monitoring systems are capable of providing real-time information over extensive areas [11]. The National Aeronautics and Space Administration (NASA), through its Fire Information for Resource Management System (FIRMS), offers global wildfire data derived from polar-orbiting satellites, including the Moderate Resolution Imaging Spectroradiometer (MODIS) and Visible Infrared Imaging Radiometer Suite (VIIRS) [12]. Although polar-orbiting satellites deliver high spatial resolution globally, their lengthy revisit intervals limit their utility for early and real-time detection [13].

In contrast, geostationary satellites offer information on target areas within short revisit times but their lower spatial resolution impedes the detection of wildfires with faint optical signals [14]. In East Asia, forest fire (FF) products from the Himawari-8 Advanced Himawari Imager (AHI) and the GEO-KOMPSAT-2A (GK-2A) Advanced Meteorological Imager (AMI) provide updates every 10 min at a spatial resolution of 2 km.

Satellite-based wildfire monitoring utilizes the thermally sensitive mid-wave infrared (MWIR) and long-wave infrared (LWIR) bands [15], [16], [17], [18]. According to Planck's law and Wien's displacement law, the wavelength at which the radiant energy from the blackbody peaks is inversely proportional to its absolute temperature [19]. Consequently, flaming areas exhibit significant brightness temperature (BT)

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differences in the MWIR and LWIR bands compared to the surrounding areas [15], [16].

Building on this theory, early stages of wildfire remote sensing utilized fixed threshold algorithms based on single (MWIR) and bispectral (MWIR–LWIR) features. These algorithms are adept at detecting wildfires with sufficiently high temperatures; however, they fail to consider the ambient baseline temperature, which complicates spatiotemporal generalization. Consequently, their accuracy is restricted to specific times and regions [19], [20], [21], [22].

To address these limitations, contextual algorithms that calculate the BT difference between the target pixel and a surrounding window have been developed, enabling early detection and improving generalization [23], [24]. Currently, all satellite-based wildfire detection products employ contextual algorithms. These algorithms presuppose homogeneous environmental conditions within the surrounding window and identify outliers by using the average BT of this window as the baseline for the target pixel [22], [25], [26], [27]. Consequently, when the environmental conditions of the target pixel differ significantly from those of the surrounding area (heterogeneous conditions), the detection performance can deteriorate. Additionally, the reliance on vector-based statistics in contextual algorithms omits the use of structural information from the image, which represents a critical capability of satellite remote sensing.

The BT values in the MWIR and LWIR bands, commonly used for satellite-based wildfire monitoring, are influenced by various environmental factors beyond just wildfires. Owing to the temperature lapse rate effect, BT is inversely related to altitude [28]. An increase in the normalized difference vegetation index (NDVI) leads to higher thermal emissivity [29], [30], which lowers surface temperatures through enhanced evapotranspiration and increased thermal inertia [31], [32], [33]. Park et al. [28] enhanced the performance of a contextual algorithm by minimizing the variability of ambient pixels using Himawari-8 AHI satellite images; however, the study primarily considered the BT lapse rate and did not adequately analyze the conditions affecting detection performance.

Several studies have utilized convolutional neural network (CNN)-based machine learning approaches to leverage the structural information of images in wildfire detection [13], [34], [35], [36], [37]. Notably, Kang et al. [35] assessed the efficacy of random forest (RF)—a tree-based model—and CNN—an image-based model—within the Korean Peninsula, and concluded that incorporating structural information of images enhances the performance of wildfire detection. However, CNNs are constrained by the necessity for a sufficiently large input window to ensure a valid pixel count, and their substantial computational demands render them less viable for real-time detection algorithms. Moreover, the effectiveness of tree-based models can be improved by including variables that capture structural information [35].

In this study, to address two major deficiencies of the conventional contextual (CC) algorithm—the neglect of environmental factors and the underutilization of structural image information—a machine learning-based wildfire detection model incorporating two advanced contextual features was developed. The detection efficacy of this model is evaluated against that of a model utilizing CC features. Additionally, an environmental correlation analysis was performed to pinpoint the conditions under which CC features falter. Furthermore, confidence and time series analyses were undertaken to ascertain potential enhancements in detection performance through the integration of each advanced contextual feature.

II. Study Area and Data

A. Study Area

The wildfire detection model discussed herein was devised and validated in South Korea, where data on wildfire occurrences is provided by the Korea Forest Service (KFS) [Fig. 1(b)]. The forest composition of the country is predominantly coniferous, highly dense, and continuous, rendering it particularly vulnerable to wildfires [38]. Additionally, South Korea's dry climate and prevailing westerly winds during winter and spring frequently culminate in extensive wildfires, thereby inflicting substantial damage [39]. The complex topography of the nation features a wide range of altitudes (with an average and standard deviation of 224 and 235 m, respectively) and a diverse vegetative cover comprising forests, crops, savannas, and grasslands.

The transferability of the wildfire detection model was evaluated using cases from Southwestern China (SWC), as reported by Liu et al. [40] [Fig. 1(c)]. This region exhibits a diverse range of vegetation and is experiencing a rapid annual increase in fractional vegetation cover [41]. According to Liu et al. [40], SWC is characterized by a subtropical monsoon climate with low precipitation levels and a dry spring season, conditions that predispose it to significant wildfire risks. The average altitude in this area is notably high at 2498 m with a substantial variation (standard deviation of 1446 m). Although similar to South Korea in terms of complex vegetation distribution, the predominant vegetation types in SWC differ slightly, including savannas, grasslands, forests, and croplands.

B. Data

1) GEO-KOMPSAT-2A AMI: The GK-2A AMI Level 1B data served as the primary input source. The AMI captures data across 16 spectral channels, ranging from the visible to the LWIR, covering both regional (East Asia) and local (Korean Peninsula) scales. In this analysis, the MWIR channel (central wavelength of 3.8 μm) and two LWIR channels (central wavelengths of 9.6 and 11.2 μm) were selected to generate input features for the model (Table I).

The MWIR and LWIR channels are influenced by both reflection and emission, with reflection effects becoming more pronounced at shorter wavelengths. To mitigate false alarms triggered by reflection effects in the MWIR channel during daylight, which have been documented in several studies [42], [43], [44], two visible channels (central wavelengths of 0.64 and 0.86 μm) were employed to reduce the incidence of false positives (FP).

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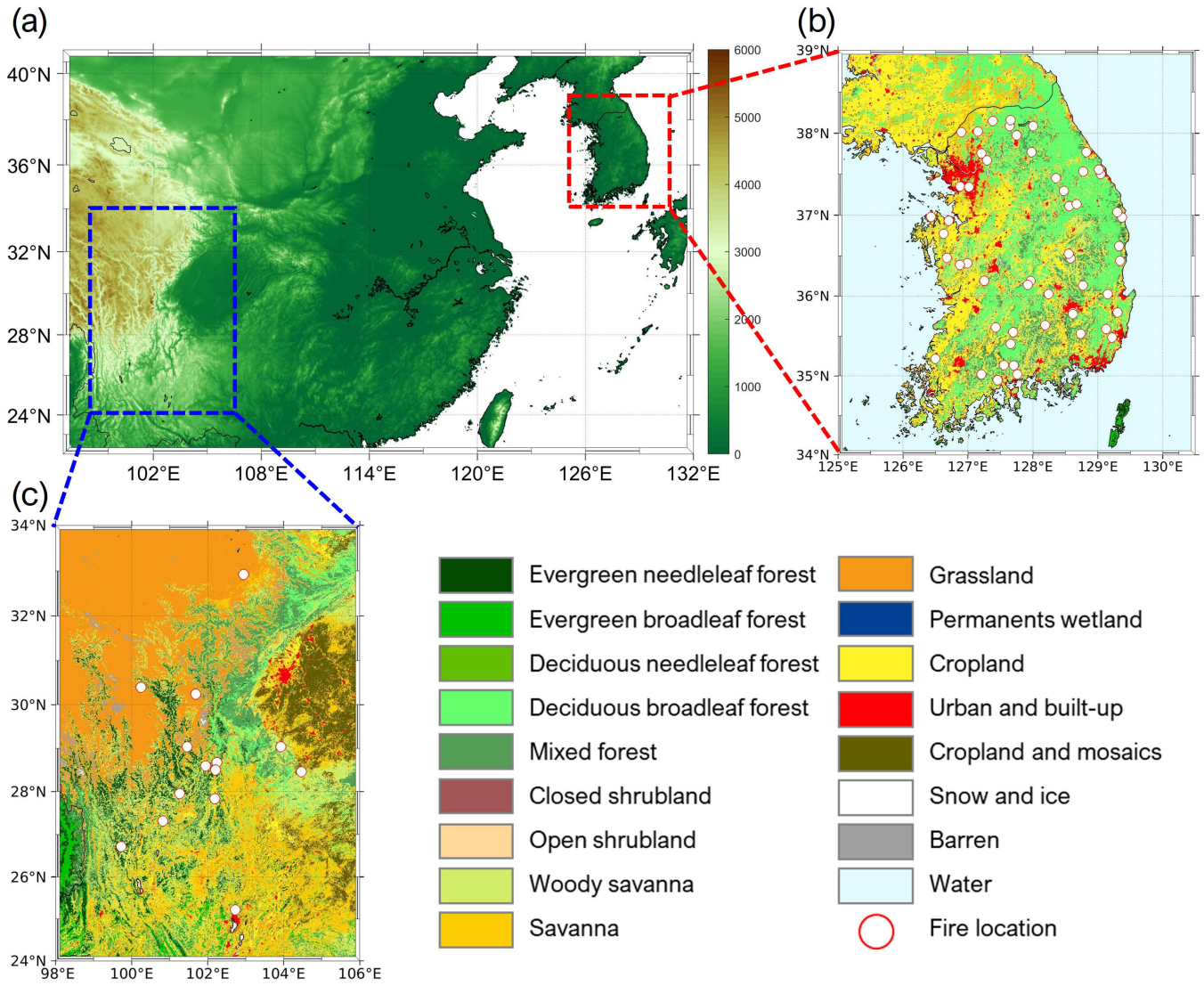


Fig. 1. Entire study area (East Asia), including the main study site (South Korea, marked with a red dotted box) for model construction and the extended study site (SWC, marked with a blue dotted box) for transferability test. (a) Elevation map from the NASA digital elevation model product (NASADEM_HGT) and (b) and (c) land cover map from the MODIS land cover type product (MCD12Q1). The fire locations shown in (b) and (c) are from the KFS and Liu et al. [40], respectively.

AMI Level 2 cloud mask data was utilized in the preprocessing stage to define clear-sky regions. The AMI cloud mask classifies images into three categories—cloud, probably cloud, and clear—based on 15 threshold tests that vary by day/night and land/sea distinctions. For this study, only the regions categorized as clear were considered clear sky. Additionally, AMI Level 2 NDVI data was employed to account for variations in BT emissivity associated with vegetation. The NDVI values range from 0 to 1, where a value closer to one indicates healthier vegetation. The AMI NDVI was calculated using normalized reflectance data with a bidirectional reflectance distribution function (BRDF) modeled for solar directional reduction. AMI Level 2 FF data, which classifies outputs into binary categories (fire and nonfire) using a MODIS-based contextual algorithm with an additional BT lapse rate correction, was employed to evaluate the detection performance of the wildfire model discussed in this study [23]. Further details on the data and retrieval algorithms are available from the National Meteorological Satellite Center (https://nmsc.kma.go.kr/enhome/html/main/main.do).

2) MODIS Land Cover Type: MODIS Level 3 land cover type data (MCD12Q1 version 6.1) was utilized to identify areas of combustible vegetation. MCD12Q1 categorizes land into 17 International Geosphere-Biosphere Program classes (11 natural vegetation, three developed and mosaic lands, and three nonvegetated lands), using a decision tree approach [45].

This study focused on 11 classes associated with forests, shrublands, savannas, grasslands, and croplands as combustible vegetation. Annual MCD12Q1 data from 2019 to 2022, featuring a global spatial resolution of 500 m, was used. Access to MCD12Q1 data was through the United States Geological Survey Earth Data (https://search.earthdata.nasa.gov/).

3) NASA Digital Elevation Model: The NASA digital elevation model (NASADEM_HGT version 1) was employed to determine the BT lapse rate relative to altitude. NASA-

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DEM, which merges reprocessed Shuttle Radar Topography Mission (SRTM) data with data from the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) and the Geoscience Laser Altimeter System (GLAS), offers improvements over its predecessors, including the resolution of voids in the SRTM digital elevation model (DEM) [46], [47], [48]. NASADEM data for February 2000, with a global resolution of 30 m, was obtained from the Land Processes Distributed Active Archive Center of the NASA Earth Observation System Data and Information Systems (https://www.earthdata.nasa.gov/eosdis/daacs).

4) KFS Fire Occurrence Data: Field observation data from the KFS was integral to constructing a fire reference database. Collected by experts through on-site visits postwildfire, this dataset includes details on ignition and extinction times, specific locations (address-based), and areas affected. The analysis covered 62 noncloud-contaminated wildfires recorded between July 2019 and July 2022, all of which resulted in damage of at least 1 hectare. Among these, the majority were minor fires damaging less than 10 hectares, while 13 were significant events affecting over 100 hectares. The most extensive wildfire, which damaged 16 302 hectares, occurred in Uljin on March 4, 2022, and persisted for approximately ten days. Additional information on these field observations is detailed in Table S1 in the supplementary material.

5) Fire Occurrence Data in Southwestern China: To mitigate validation bias toward specific datasets and locations, additional fire occurrence data was acquired from Southwestern China for the transferability test. Liu et al. [40] documented the ignition dates and locations (coordinates based on latitude and longitude) for 14 wildfire incidents that occurred in the region from 2020 to 2022. Among these, 13 cases that coincided with the study period were selected for analysis. Further details on these data can be found in Liu et al. [40] and Table S2 in the supplementary material.

III. Methods

A research flow diagram for this study is depicted in Fig. 2. The process involved transforming satellite data into four types of input features through data preprocessing and input feature construction steps. The field observation data was formatted to align with the same spatiotemporal resolution as that of the GK-2A AMI during the reference construction phase. In the wildfire detection phase, the likelihood of fire occurrence was determined using a machine learning-based classification model. This was followed by a false alarm rejection step, where the final wildfire product was output as a binary value indicating either fire or nonfire conditions.

A. Data Preprocessing

In this study, given that the wildfire detection products are required to match the spatiotemporal resolution of the GK-2A AMI, all input satellite data were resampled. Spatially, the data was resampled to a resolution of 2 km using bilinear interpolation for continuous variables (VI06, VI08, and DEM) and majority voting for discrete variables (LC). Temporally, the resampling involved aligning data to the closest acquisition time of the GK-2A AMI, occurring every 2 min.

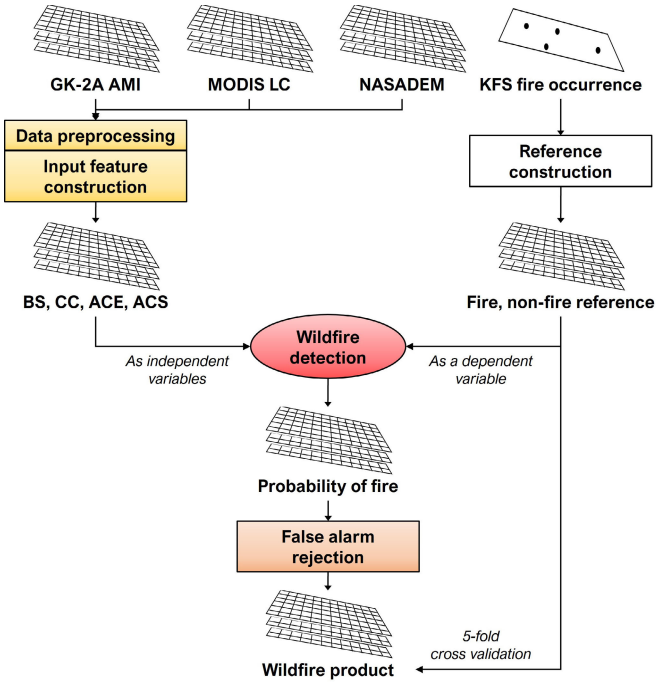


Fig. 2. Research flow diagram. BS, bispectral feature; CC, conventional contextual feature; ACE, advanced contextual feature considering baseline environment; and ACS, advanced contextual feature considering structure similarity.

The wildfire detection model developed in this study was designed to target terrestrial regions characterized by combustible vegetation and clear skies. Nonvegetated areas such as urban, barren, and water bodies often display high BT values during the daytime, not due to wildfires but because of their high reflectivity [42], [43], [44].

Additionally, BT signals from the ground may be obscured by clouds, leading to lower readings than those in surrounding areas. These factors complicate the accurate computation of contextual features and degrade the performance of the model. Consequently, areas that are nonvegetated or cloud-covered were systematically excluded from the outputs of the model.

B. Input Feature Construction

Four types of input features were employed in the machine learning-based wildfire detection model, as detailed in Table II. Conventionally, bispectral (BS) and CC features have been utilized for wildfire detection. This study introduced two advanced contextual features designed to address the limitations of the CC algorithm: the advanced contextual feature considering the baseline environment (ACE) and the advanced contextual feature for structural similarity (ACS). ACE and ACS are indices that represent spatial anomalies and spatial similarities, respectively, and include adjustments for environmental detrending and measurements of structural similarity.

To evaluate the performance of these features, three experimental schemes were implemented: Scheme 1 utilized BS and CC features; Scheme 2 utilized BS and ACE features; and Scheme 3 utilized BS, ACE, and ACS features. Additionally, to isolate the effects of DEM and NDVI detrending, two further schemes were established (refer to Section V): Scheme

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TABLE I
LIST OF SATELLITE DATA SOURCES USED IN THIS STUDY

Source	Abbreviation	Description	Spatial resolution	Temporal resolution
GK-2A AMI	VI06	Reflectance at center wavelengths 0.64 and 0.86 μm	500 m	Regional area: 10 min Local area: 2 min
	VI08		1 km	
	SW38	Brightness temperature at center wavelengths 3.83, 9.62, and 11.23 μm	2 km	
	IR96			
	IR112			
	CLD	AMI Level 2 cloud mask product		
	NDVI	AMI Level 2 normalized difference vegetation index product		1 day
	FF	AMI Level 2 forest fire product		10 min
Terra, Aqua MODIS	LC	MODIS Level 3 land cover type product	500 m	1 year
SRTM, ASTER, GLAS	DEM	Multi-satellite digital elevation model product	30 m	–

TABLE II
LIST OF INPUT FEATURES USED IN WILDFIRE DETECTION MODEL

Type	Abbreviation	Description
Bi-spectral feature (BS)	SW38-IR96	Spectral difference between brightness temperature at center wavelengths 3.8 and 9.8 μm
	SW38-IR112	Spectral difference between brightness temperature at center wavelengths 3.8 and 11.2 μm
Conventional contextual feature (CC)	$\text{SW38-IR96}_{\text{CC}}$	Contextual difference between SW38-IR96 of the target and surrounding window
	$\text{SW38-IR112}_{\text{CC}}$	Contextual difference between SW38-IR112 of the target and surrounding window
Advanced contextual feature considering baseline environment (ACE)	$\text{SW38-IR96}_{\text{ACE}}$	Environmentally detrended $\text{SW38-IR96}_{\text{CC}}$
	$\text{SW38-IR112}_{\text{ACE}}$	Environmentally detrended $\text{SW38-IR112}_{\text{CC}}$
Advanced contextual feature considering structure similarity (ACS)	$\text{SW38-IR96}_{\text{ACS}}$	Structural similarity between the reference fire image and input image of $\text{SW38-IR96}_{\text{ACE}}$
	$\text{SW38-IR112}_{\text{ACS}}$	Structural similarity between the reference fire image and input image of $\text{SW38-IR112}_{\text{ACE}}$

2-1 employed BS and ACE with only DEM detrending and Scheme 2-2 used BS and ACE with only NDVI detrending.

In this study, the BT difference between the MWIR (SW38) and two LWIR (IR96 and IR112) channels was defined as a BS feature (Fig. 3). CC features were calculated by measuring the contextual difference on the BS, which involves computing the difference between the BT of the target pixel and the mean BT of the valid pixels in the surrounding window (Fig. 3). Here, valid pixels are those within the window that are not adjacent to the target pixel, are not affected by clouds, and are vegetated.

The initial size of the surrounding window was set to 7×7, centered on the target pixel. If the percentage of valid pixels within this window was $\leq 25\%$, the window size was incrementally increased to a maximum of 15×15 to ensure sufficient data for accurate calculation. The BS and CC features were calculated using (1) and (2), respectively

$$\text{BS} = \text{MWIR}_T - \text{LWIR}_T \tag{1}$$

$$\text{CC} = \text{BS}_T - \overline{\text{BS}}_V \tag{2}$$

where MWIR_T, LWIR_T, and BS_T denote the BT of the MWIR and LWIR channels and the BS value of the target pixel,

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respectively, and $\overline{\mathrm{BS}}_V$ denotes the average BS value of the valid pixels in the surrounding window.

The ACE was calculated by performing environmental detrending alongside contextual difference measurements on BS (Figs. 3 and 4). Environmental detrending involves the removal of the influences of DEM and NDVI from BS and consists of the following three steps.

1) Pixel pair extraction: To minimize the spatiotemporal dependence of the environmental correlation, 100 pixel pairs were extracted from adjacent locations every hour during the study period.
2) Environmental correlation training: The entire set of pixel pairs was divided into subsets based on the solar zenith angle (SOZ) in 10° increments (from 10° to 170°), and the correlation between the environmental factors and BS for each subset was trained using a multilinear regression (MLR) model. The environmental correlations for each SOZ range are presented in Figs. S1 and S2 (see the supplementary material).
3) Environmental correlation detrending: Based on the trained correlation, the environmental conditions within the surrounding window were adjusted to the same level as the target pixel, and the influences of DEM and NDVI were removed from the BS.

The process of environmental correlation detrending is shown in the following equation, which was performed on valid pixels in the surrounding window before calculating the contextual difference:

$$\mathrm{BS}_{\mathrm{VE}} = \mathrm{BS}_V - a(\mathrm{DEM}_V - \mathrm{DEM}_T) - b(\mathrm{NDVI}_V - \mathrm{NDVI}_T) \tag{3}$$

where $\mathrm{BS}_V, \mathrm{DEM}_V, \mathrm{DEM}_T, \mathrm{NDVI}_V$, and NDVI_T denote the BS, DEM, and NDVI values for the valid and target pixels, respectively. Coefficients a and b represent the time-dependent correlations between the environmental factors and the BS trained using the MLR model and $\mathrm{BS}_{\mathrm{VE}}$ denotes the BS value of the valid pixels after environmental detrending.

The ACS was calculated by performing environmental detrending alongside structural similarity measurements on BS (Figs. 3 and 4). Structural similarity measurement aims to evaluate the spatial similarity between the input and reference fire image and consists of the following three steps.

1) Input image generation: The invalid pixel values in the detrended BS were replaced with the average values of the valid pixels. These values were then normalized to a range of 0–1 using min–max scaling to generate the input image.
2) Reference fire image creation: A reference fire image, matching the size of the input image, was generated. This image represents a typical wildfire pattern where the highest BT signal is located at the target pixel, and it decreases with distance from the target, thereby reflecting the common characteristics of wildfire images [Fig. 4(b)].
3) Structural similarity index measurement (SSIM): An SSIM map between the input and the reference fire images was generated to assess their structural similarity. The SSIM value for the target pixel was then assigned to the ACS.

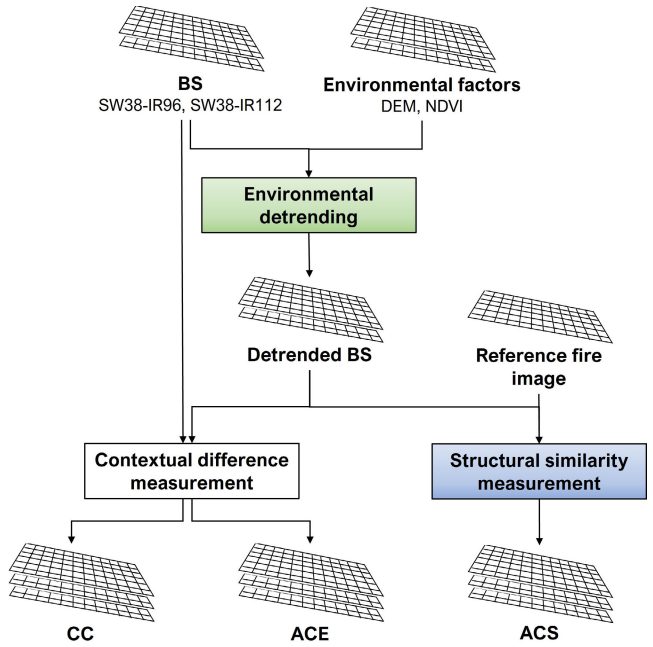


Fig. 3. Research flow diagram of the input feature construction step. BS, bispectral feature; CC, conventional contextual feature; ACE, advanced contextual feature considering baseline environment; and ACS, advanced contextual feature considering structure similarity.

The SSIM is a perceptual metric that measures image quality by comparing local patterns of pixel intensities, which are normalized for luminance, contrast, and structural integrity [49]. The local SSIM for each pixel was calculated using the following equation:

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \tag{4}$$

where x and y denote the local circular–symmetric windows of the input and reference fire image, respectively, and $\mu_x, \mu_y, \sigma_x, \sigma_y$, and σ_{xy} represent the circular–symmetric Gaussian weighted mean, standard deviation, and covariance of x and y, respectively [49]. C_1 and C_2 can be calculated as $C_1 = (0.01 \times L)^2$ and $C_2 = (0.03 \times L)^2$, where L represents the dynamic range of the input image. As the input image normalized to the range of 0–1 was used in this study, the value of L was 1.

C. Reference Construction

The KFS fire occurrence data provides address-based information on the locations of wildfires. Therefore, geocoding was utilized to translate the address information into geographical coordinates (latitude and longitude) and subsequently aligned with the GK-2A AMI grid. Note that the matched locations might not precisely correspond to the actual wildfire locations, and the location of the active fire may shift over time within the same wildfire case. To address these potential inaccuracies, three corrective steps were implemented to derive fire references that accurately reflect the real-time fire locations at the same spatiotemporal resolution as the GK-2A AMI.

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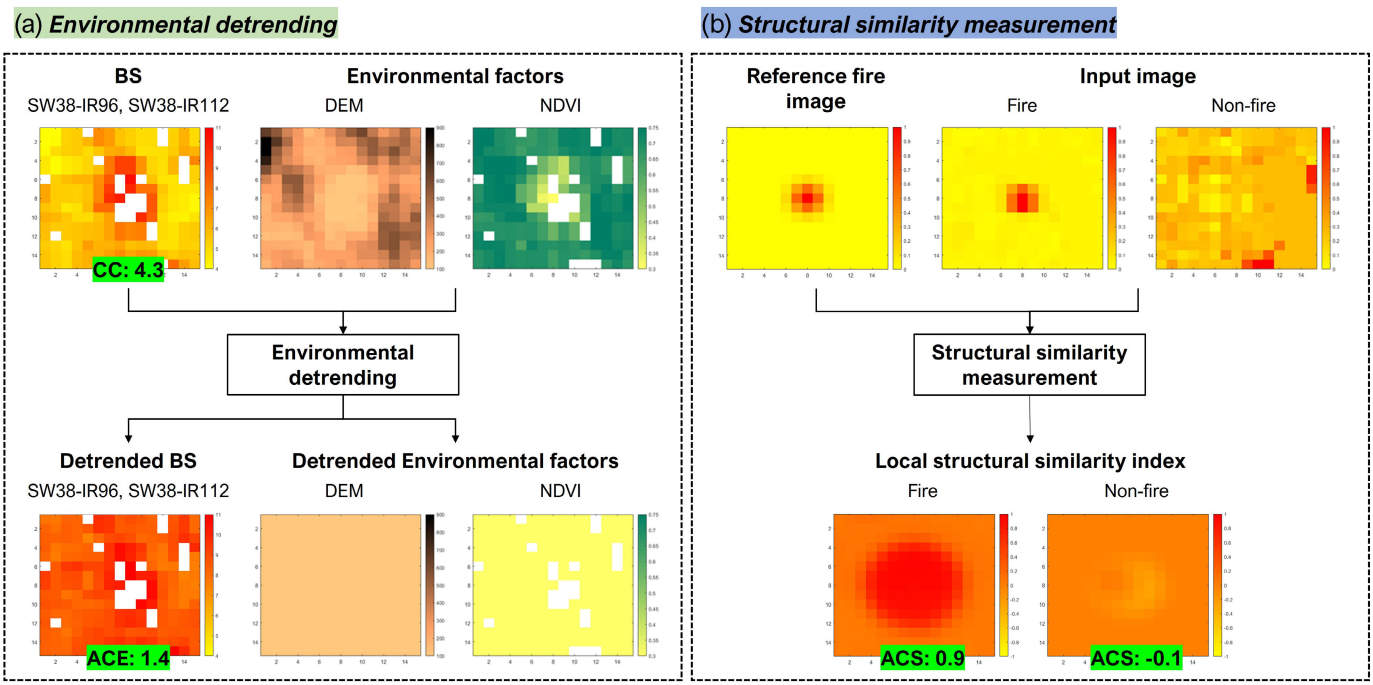


Fig. 4. Flow diagram of the environmental detrending and structural similarity measurement. (a) Spatial anomaly index (i.e., CC and ACE) of the nonfire sample decreased after the environmental detrending. (b) Prominent distinction observed between the spatial similarity index (i.e., ACS) of the fire and nonfire samples. BS, bispectral feature; CC, conventional contextual feature; ACE, advanced contextual feature considering baseline environment; and ACS, advanced contextual feature considering structure similarity.

1) Spatial relocation: To account for potential spatial discrepancies between the address-based locations and the GK-2A AMI grid, fire references were extracted within a 5 × 5 window centered on the KFS fire occurrence location.
2) Extraction with real-time statistics: For identifying dynamically changing wildfire locations, pixels where the BS value exceeded the Q3 (upper quartile) + 3 × interquartile range (IQR) within the surrounding window for each timestamp were selected.
3) Extraction with time series statistics: To identify areas impacted by wildfires over the duration of the fire, pixels where the BS value consistently exceeded Q3 + 3 × IQR within the surrounding window for more than a quarter of the fire duration were selected.

Only pixels fulfilling all three criteria were used as fire references, enhancing the accuracy of the reference data at the same spatiotemporal resolution as the satellite imagery using in situ data.

To construct nonfire references, which are essential for calibrating the wildfire detection model under various environmental conditions, three additional steps were executed.

1) Nonfire candidate selection: To unequivocally eliminate any influence of wildfires on the contextual feature calculations of nonfire references, a 15 × 15 window centered on the KFS fire occurrence location was omitted from the nonfire sample extraction process.
2) Extraction with environmental representation: Nonfire samples were extracted based on each percentile range of DEM and NDVI values to ensure that these references are environmentally indicative.
3) Extraction with spatiotemporal representation: Nonfire samples were systematically gathered across the study area at consistent hourly intervals to ensure spatiotemporal representativeness.

Ultimately, 31 687 fire and 233 737 nonfire samples were compiled to construct and validate the wildfire detection model. This extensive dataset supports robust model training and testing, ensuring the effectiveness of the model across varying conditions and enhancing its predictive accuracy.

D. Wildfire Detection

In the wildfire detection step, three widely used machine learning models were evaluated: the light gradient boosting machine (LGBM), RF, and multilayer perceptron (MLP). LGBM and RF are ensemble models based on decision trees, utilizing boosting- and bagging-based techniques, respectively. Boosting sequentially trains new base learners by leveraging the gradient from previous learners to minimize residual errors [50], [51], whereas bagging trains each base learner independently on randomly selected samples and features using bootstrap aggregating to minimize overfitting [52].

The MLP is a neural network model with layers of perceptrons (input, multiple hidden, and output layers) that simulates human neural processing [53]. The connections between perceptrons, characterized by weights and biases, are refined through feed-forward and backpropagation methods to optimize model performance [54]. All three models were implemented using the PyCaret package in Python, and the hyperparameters were optimized using a random grid search.

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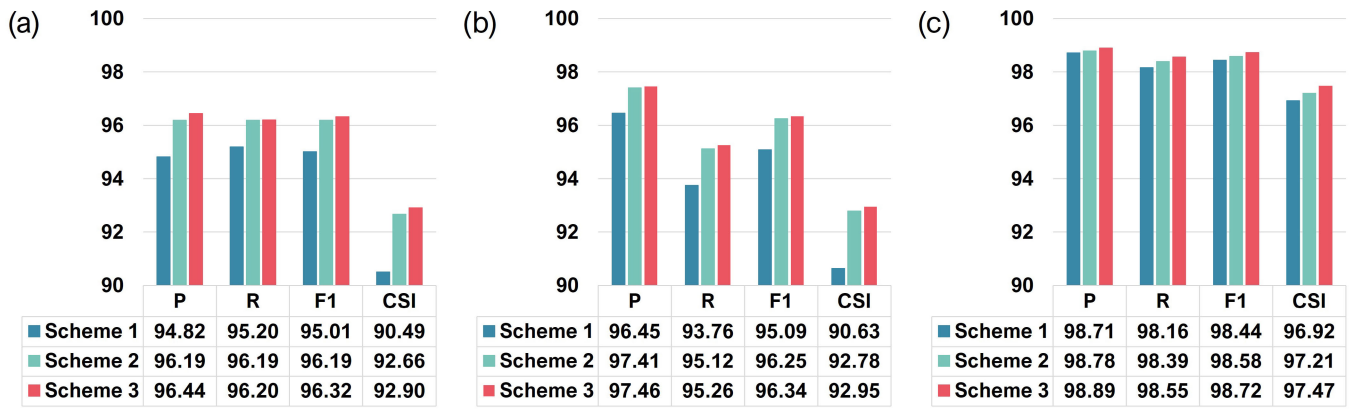


Fig. 5. Accuracy evaluation results of the three comparative schemes of the LGBM for (a) daytime before false alarm rejection, (b) daytime after false alarm rejection, and (c) nighttime. P, precision; R, recall; $F1$, $F1$-score; and CSI, critical success index.

E. False Alarm Rejection

The results from the wildfire detection model employing the MWIR and LWIR bands may include FP, particularly due to reflectance effects in highly reflective areas during the daytime [42], [43], [44], [55]. To mitigate these inaccuracies, a false alarm rejection process was employed, which utilized a threshold algorithm based on the reflectance values from two visible bands (VI06 and VI08) of the GK-2A AMI [26], [44].

Pixels with a high probability of fire (>0.9) were directly classified as wildfires, which bypassed the rejection process. Conversely, pixels with a moderate probability of fire (0.5–0.9) were only classified as wildfires if they satisfied the threshold criteria in the rejection process. These thresholds were determined through statistical analyses of the outputs of the model which were set at 0.1 for VI06, 0.2 for VI08, and 0.1 for contextual differences in each band.

F. Evaluation and Comparison Methods

Wildfire detection results were validated using fivefold cross validation on 62 cases provided by the KFS in South Korea. The environmental conditions influencing wildfire ignition vary spatially and temporally [56], [57]. To account for the impact of these conditions, each validation fold was stratified by ignition date and case number.

The fire reference used in the reference construction phase was developed utilizing the GK-2A AMI at locations corresponding to KFS fire occurrence data. For the transferability test, however, fire references were constructed using data from the MODIS and VIIRS satellites. Active fire products from MODIS and VIIRS, identified within three days of the ignition date and within a 15 × 15 window centered on the wildfire location, served as the basis for fire references.

Fire and nonfire events were encoded as 1 and 0, respectively, allowing the use of a confusion matrix to illustrate the actual observations versus model predictions in a 2 × 2 matrix format. This matrix categorizes predictions as true positives (TP), false negatives (FN), FP, and true negatives (TN). Among the performance metrics derived from the confusion matrix, four metrics less sensitive to class imbalance were employed: precision, recall, $F1$-score, and the critical success index (CSI). These metrics were computed as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{5}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{6}$$

$$F1\text{-score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{7}$$

$$\text{Critical success index} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{8}$$

IV. Results

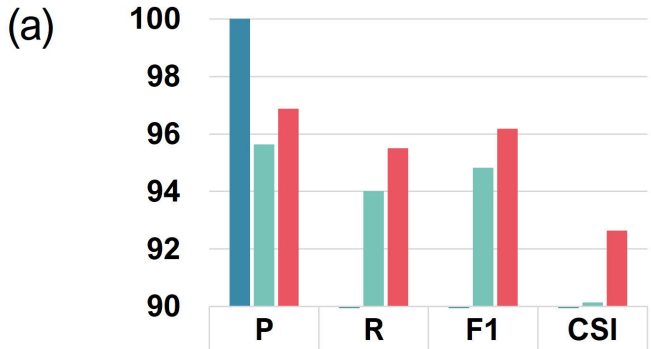
A. Comparative Accuracy Evaluation

The effectiveness of the three comparative schemes for each machine learning model was assessed to evaluate the impact of advanced contextual features relative to CC features (Figs. 5, S3, and S4). The performance disparities between the models were minimal, with all schemes achieving $F1$-scores above 90%, indicating robust detection capabilities of the machine learning-based wildfire detection models. Furthermore, the consistent trend of performance improvement across all schemes and models suggests that the advanced contextual features (ACE and ACS) are versatile and enhance performance across different machine learning algorithms.

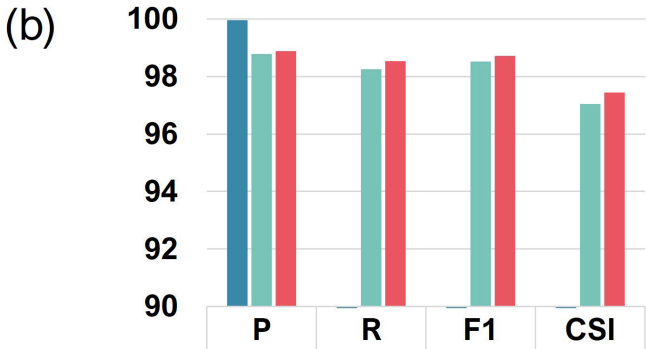
In general, the nighttime results exhibited higher performance compared to the daytime. This improvement is attributed to the absence of solar radiation at night, eliminating reflectance effects on BT, thus creating more homogeneous conditions and a lower baseline temperature. During the day, the application of the false alarm rejection process marginally reduced recall but increased precision significantly, which maintained the $F1$-score and CSI nearly unchanged. Given the comparable performance across the three models, further analyses focused on the LGBM, owing to its significantly faster computational speed.

The evaluation of the detection performance across different schemes indicated that both the $F1$-score and CSI progressively improved from Scheme 1 to Scheme 3. Compared to Scheme 1, which used CC features, Scheme 2, which

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	P	R	F1	CSI
GK-2A forest fire	100.00	38.76	55.86	38.76
LGBM Scheme 1	95.61	94.00	94.80	90.11
LGBM Scheme 3	96.85	95.48	96.16	92.61



	P	R	F1	CSI
GK-2A forest fire	99.93	56.94	72.55	56.92
LGBM Scheme 1	98.76	98.23	98.49	97.03
LGBM Scheme 3	98.86	98.52	98.69	97.41

Fig. 6. Accuracy evaluation results of the GK-2A FF products and two comparative schemes of the LGBM for (a) daytime after false alarm rejection and (b) nighttime. P, precision; R, recall; $F1$, $F1$-score; and CSI, critical success index.

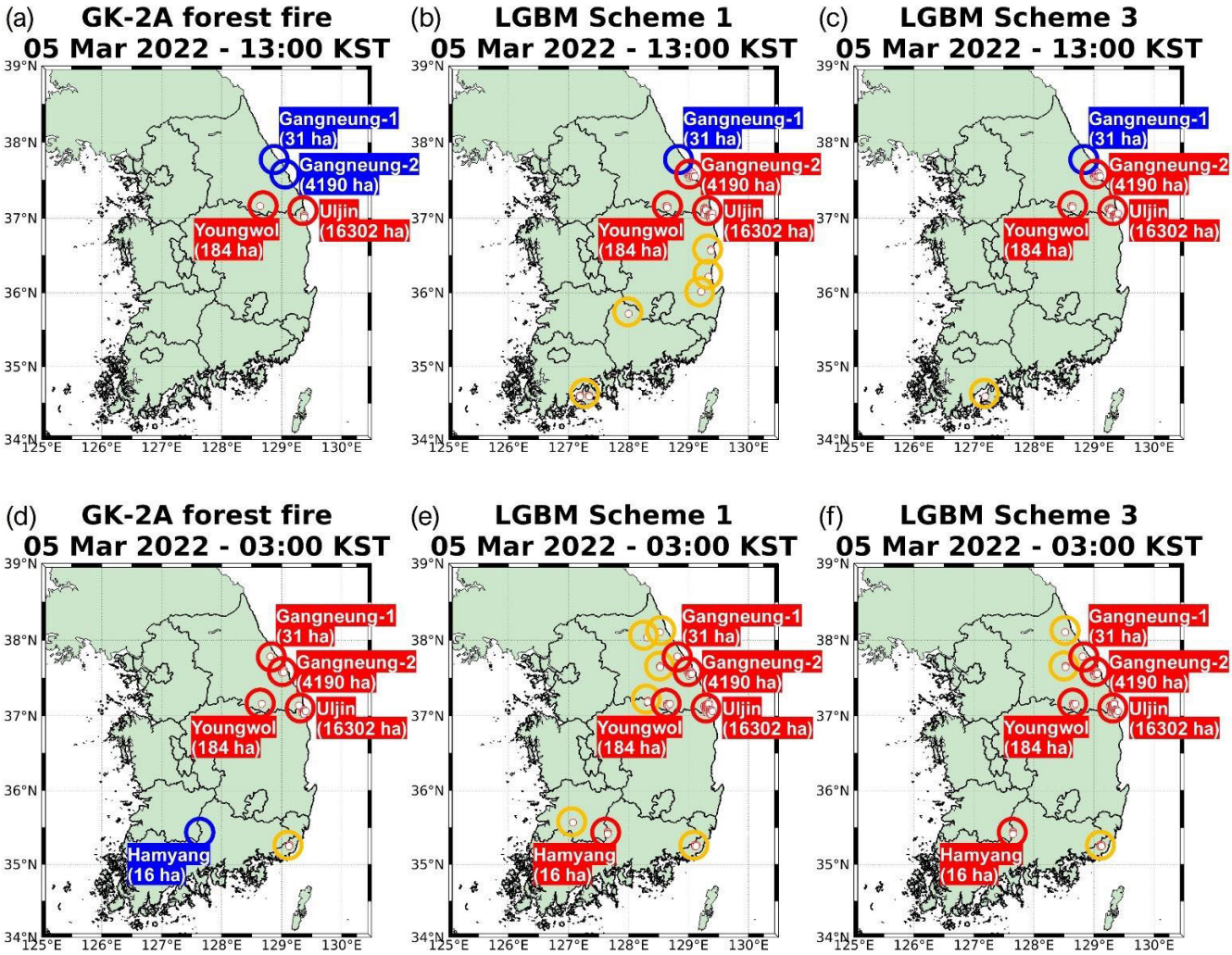


Fig. 7. Spatial distribution of wildfire detection results from GK-2A FF products and two comparative schemes of the LGBM for (a)–(c) daytime and (d)–(f) nighttime. TP, red circles; FN, blue circles; and FP, orange circles.

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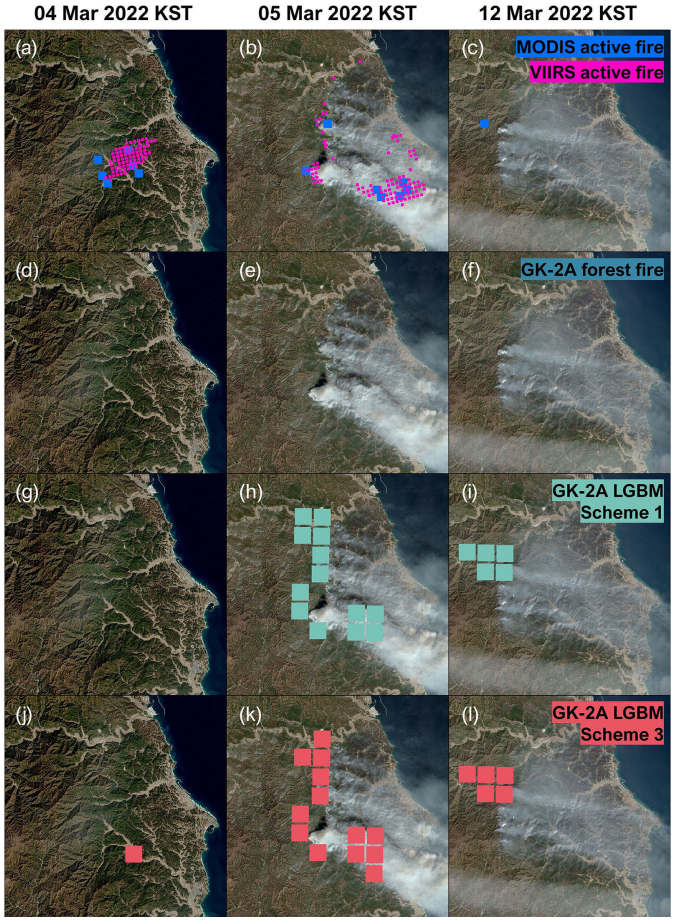


Fig. 8. Spatial distribution of wildfire detection results from (a)–(c) MODIS and VIIRS active fire products, (d)–(f) GK-2A FF products, and (g)–(l) two comparative schemes of the LGBM for the Uljin case. The background images are red, green, blue (RGB) composites of the Sentinel-2 Level 2A surface reflectance product (S2MSI2A) from (a), (d), (g), and (j) February 23, (b), (e), (h), and (k) March 5, and (c), (f), (i), and (l) March 10, 2020, respectively.

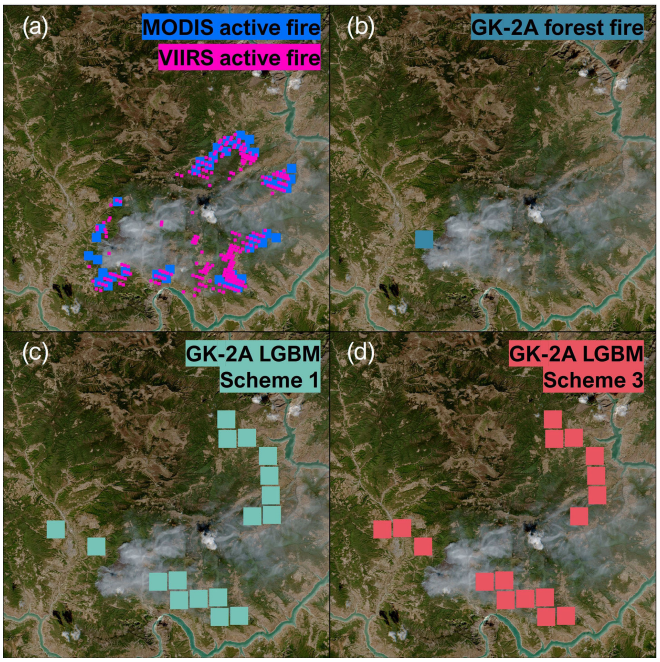


Fig. 9. Spatial distribution of wildfire detection results from (a) MODIS and VIIRS active fire products, (b) GK-2A FF products, and (c) and (d) two comparative schemes of the LGBM for the SWC-1 case. The background image is an RGB composite of the Sentinel-2 Level 2A surface reflectance product (S2MSI2A) on March 30, 2020.

incorporated the ACE, demonstrated improvements in precision and recall during daytime operations. Environmental correlation analysis revealed a stronger negative correlation between environmental factors (DEM and NDVI) and BT during the day than at night (Figs. S1 and S2; refer to the supplementary material). This pronounced correlation during the day introduced BT heterogeneity, complicating the training of machine learning algorithms and diminishing overall detection performance. By substituting CC with ACE, which provides environmentally detrended contextual data, there was a notable enhancement in model performance.

Furthermore, Scheme 3, which also included the ACS, exhibited marginal improvements in recall over Scheme 2. Although ACE primarily addresses spatial anomalies related to temperature, rendering it less effective for low-intensity wildfire detection, ACS, being a spatial similarity index based on structural characteristics, remains largely independent of wildfire intensity. Detailed discussions on the environmental correlation and conditions enhancing performance through ACE and ACS are included in Section V.

To compare the wildfire detection capabilities of the GK-2A FF product and the LGBM model, which operate at different temporal resolutions (10 and 2 min, respectively), evaluations were conducted on samples re-extracted at 10 min intervals (Fig. 6). The GK-2A FF demonstrated extremely high precision but significantly low recall, especially during the daytime. In contrast, the LGBM model allowed for certain FP but detected more wildfires based on its high detection sensitivity.

B. Comparison of Spatial Distribution

The spatial distribution of the wildfire detection by the GK-2A FF and LGBM models was compared for the historic wildfire that occurred in South Korea on March 5, 2022 (Fig. 7). Both models exhibited superior detection performance during the nighttime over the daytime, which is consistent with the findings from the comparative accuracy evaluation. Notably, the GK-2A FF did not detect the Hamyang wildfire (case number 41 in Table S1; refer to the supplementary material), which was the smallest recorded during the nighttime period. In contrast, the LGBM model successfully detected all five wildfire cases, which demonstrated its high detection sensitivity.

During daytime hours, neither model detected the Gangneung-1 wildfire (case number 42 in Table S1; refer to the supplementary material), which started at 22:00 KST on March 4 and was extinguished by 21:00 KST on March 5. By 13:00 KST on March 5, the combustion had largely subsided, rendering the BT anomaly undetectable. Additionally, the GK-2A FF did not identify the Gangneung-2 wildfire (case number 43 in Table S1), the second largest case, despite observable high BT values in the area. This failure is likely attributable to the false alarm rejection process of the GK-2A

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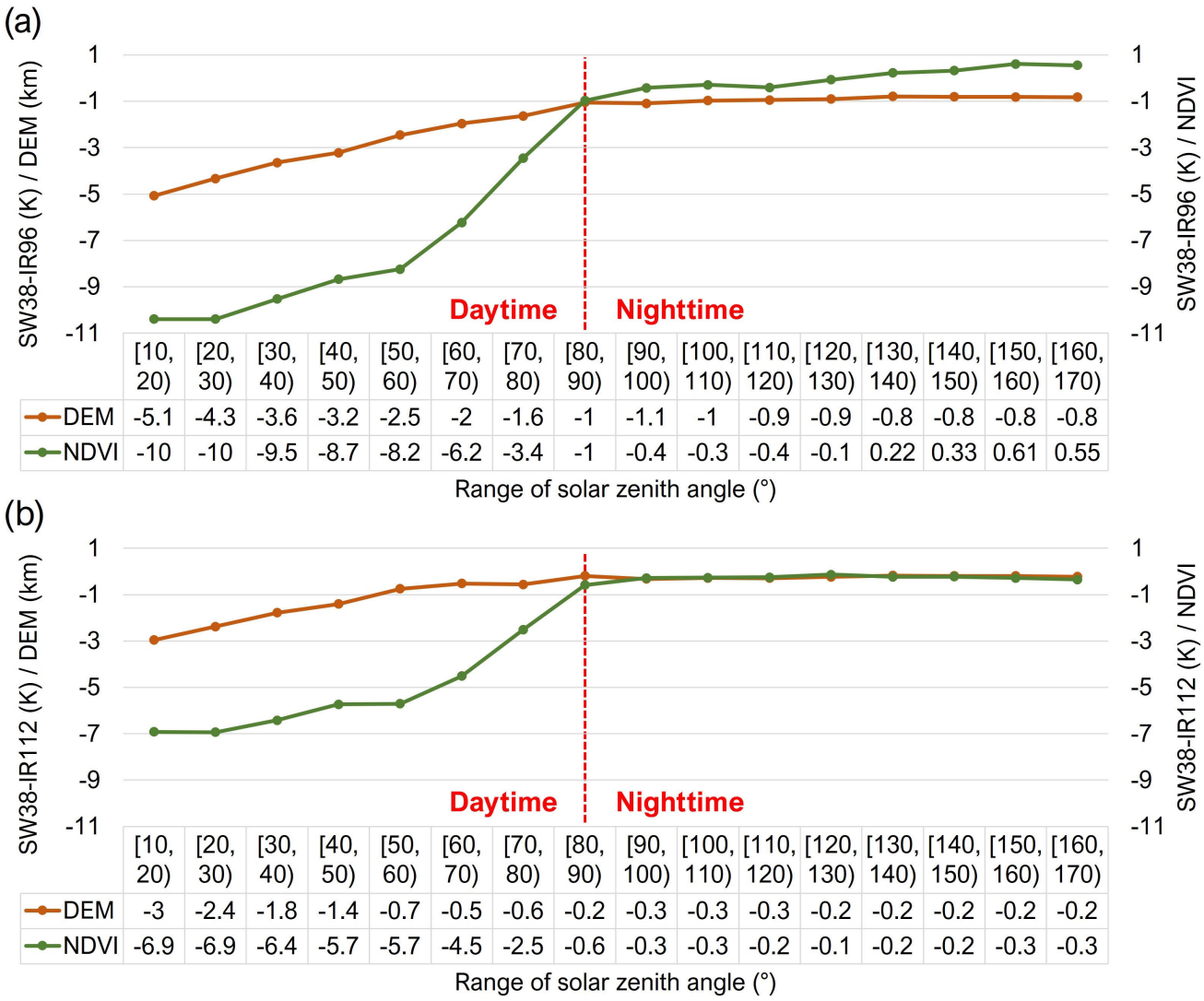


Fig. 10. Correlation plot between environmental factors and BS features. (a) SW38-IR96 and (b) SW38-IR112 for each range of SOZ.

FF, which disregards any pixel as a fire pixel if its NDVI falls below a predetermined threshold relative to its surrounding area.

Although the GK-2A FF model applies stringent thresholds and a robust false alarm rejection process to minimize FP, it occasionally misses both small and large fires. Conversely, the LGBM model, with its heightened detection sensitivity, consistently identifies smaller fires but may trigger certain false alarms. Nevertheless, the implementation of ACE and ACS in Scheme 3 effectively reduced the number of false alarms compared to Scheme 1, enhancing detection accuracy during both day and night. These improvements are corroborated by the performance metrics detailed in the comparative accuracy evaluation (Figs. 5 and 6).

For the Uljin wildfire (case number 39 in Table S1; see the supplementary material), which resulted in the largest area of damage, the LGBM model demonstrated a spatial distribution comparable to the MODIS and VIIRS active fire products, which exhibited resolutions of 1 km and 375 m, respectively, despite training the LGBM at the spatial resolution (2 km) of the GK-2A (Fig. 8).

Notably, LGBM Scheme 3 achieved its first detection at 02:20 KST on March 5, which was only a 2 min delay from the recorded ignition time by the KFS, showcasing the high early detection capability of the model at the spatiotemporal resolution of the GK-2A [Fig. 8(j)]. Both comparative schemes of the LGBM concluded their detections by 12:30 KST on March 12, limited by cloud cover which prevented further detection [Fig. 8(i) and (l)]. The GK-2A FF product failed to detect any wildfires during these times due to its low detection sensitivity and stringent false alarm rejection processes [Fig. 8(d)–(f)].

C. Transferability Test in Southwestern China

Transferability tests for Schemes 1 and 3, along with the GK-2A FF, were conducted in SWC (Table III). All three products exhibited low recall rates of less than 50%, primarily because they were operational at the spatial resolution of GK-2A (2 km), whereas the fire references used in the tests were established at the finer spatial resolutions of MODIS and VIIRS (1 km and 375 m, respectively). This discrepancy is not because of the limitations of the algorithm but due

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TABLE III

TRANSFERABILITY TEST RESULTS OF THREE COMPARATIVE WILDFIRE PRODUCTS. RECALL MODIS AND RECALL VIIRS ARE THE RECALLS CALCULATED BY ASSUMING MODIS AND VIIRS ACTIVE FIRE PRODUCTS AS A REFERENCE, RESPECTIVELY

Wildfire Products	Recall MODIS	Recall VIIRS
GK-2A forest fire	5 / 54 (9.3 %)	20 / 212 (9.4 %)
LGBM Scheme 1	20 / 54 (37.0 %)	75 / 212 (35.4 %)
LGBM Scheme 3	24 / 54 (44.4 %)	82 / 212 (38.7 %)

to the technical differences between the sensors employed. Consequently, certain smaller fires that were detectable by MODIS and VIIRS were possibly dismissed as noise by the GK-2A owing to its coarser spectral sensitivity.

The advent of geostationary satellites with higher spatial resolutions may mitigate these limitations. Therefore, the focus was on analyzing relative recall rather than absolute values in the transferability test. The GK-2A FF exhibited a very low recall of approximately 9%, whereas the LGBM model recorded relatively higher recall values between 35% and 44%. Notably, LGBM Scheme 3 displayed the greatest detection sensitivity, affirming its capability to surmount certain technical challenges through algorithmic refinements.

The spatial distribution for the SWC-1 case (case number 1 in Table S2; refer to the supplementary material) exhibited a pattern similar to that observed in South Korea (Fig. 9). However, the LGBM model generated additional FP in some nonvegetated areas. This issue was partly due to the resampling of MODIS LC data (500 m) to the GK-2A resolution (2 km), during which errors in LC classification were likely magnified. In mixed vegetation and nonvegetation areas, majority voting during the resampling process might classify a pixel as vegetated, but it could still be influenced by the surrounding nonvegetated regions, resulting in higher BT readings. These discrepancies might be reduced in the future with the use of geostationary satellite images offering higher spatial resolution.

V. DISCUSSION

A. Environmental Correlation Analysis

The fitted slope of the MLR model for each range of the SOZ used for environmental detrending was analyzed (Fig. 10). During the daytime (SOZ $<$ 85°), the smaller the SOZ, the stronger the negative correlation observed between environmental factors and BS. Conversely, at nighttime (SOZ $\geq$ 85°), this correlation nearly vanished. The SW38 band, employed in calculating the BS, is influenced predominantly by emissivity at night and by reflectance during the day [58], [59], [60]. Consequently, the correlations between daytime and nighttime conditions were markedly distinct. The temperature lapse rate relative to altitude displays a diurnal pattern, where a weaker solar warming effect governs a stronger settling of cooler air, thereby diminishing the lapse rate [61]. Given the variation in thermal inertia associated with vegetation density, the cooling effect of the NDVI on temperature intensifies under strong solar radiation [31], [62].

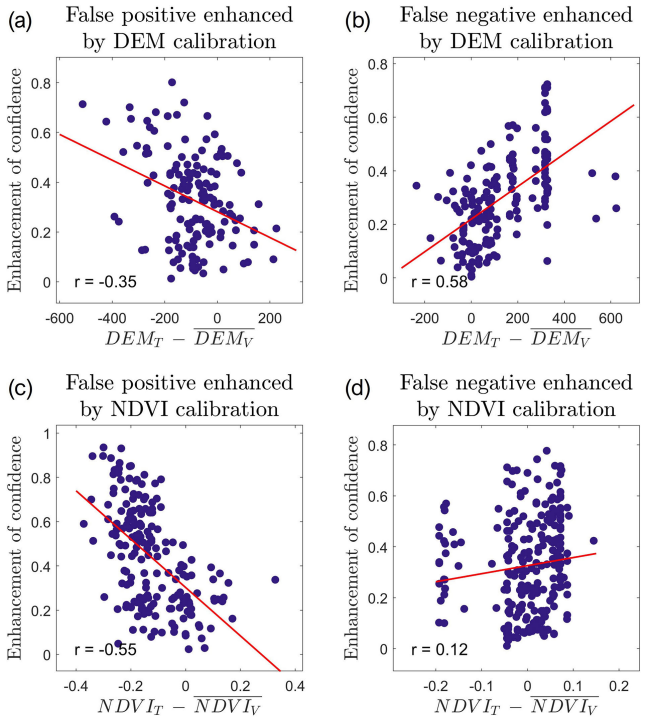


Fig. 11. Correlation plots between contextual differences in environmental factors and enhancement of confidence for (a) and (c) nonfire and (b) and (d) fire samples showing improved performance with environmental detrending. DEM_T, NDVI_T, $\overline{\mathrm{DEM}_V}$, and $\overline{\mathrm{NDVI}_V}$ indicate the DEM and NDVI of the target pixels and the average DEM and NDVI of the valid pixels in the surrounding window, respectively. Note that a larger difference in environmental factors between the target and surrounding window corresponds to a greater enhancement of confidence.

The environmental correlation analysis involving the DEM, NDVI, and radiation temperature conformed to the expected diurnal pattern (Fig. 10). During the period defined by SOZ 2 (10° $\leq$ SOZ $<$ 20°), the strongest environmental correlations were observed (correlations between DEM and BS were -5.1 and -3.0 K/km, and between NDVI and BS were -10 and -6.9 K, respectively), leading to significant temperature anomalies that were attributable solely to environmental factors without wildfire influence. These findings underscore the limitations of traditional BT anomaly-based approaches in wildfire detection and highlight the critical requirement to incorporate environmental variables into the analysis.

B. Confidence Analysis

A confidence analysis was performed to ascertain the conditions under which environmental detrending enhanced the performance of the wildfire detection model (Fig. 11). This analysis assessed the degree of confidence enhancement, defined as the extent to which the probability of fire decreased for nonfire samples (FP in Scheme 1 and TN in Schemes 2-1 and 2-2) and increased for fire samples (FN in Scheme 1 and TP in Schemes 2-1 and 2-2).

In Scheme 1, a high incidence of FP was observed when the DEM and NDVI values of the target pixel were lower than those of the surrounding window [Fig. 11(a) and (c)]. Conversely, most FN occurred under the opposite conditions [Fig. 10(b) and (d)]. As indicated in the environmental

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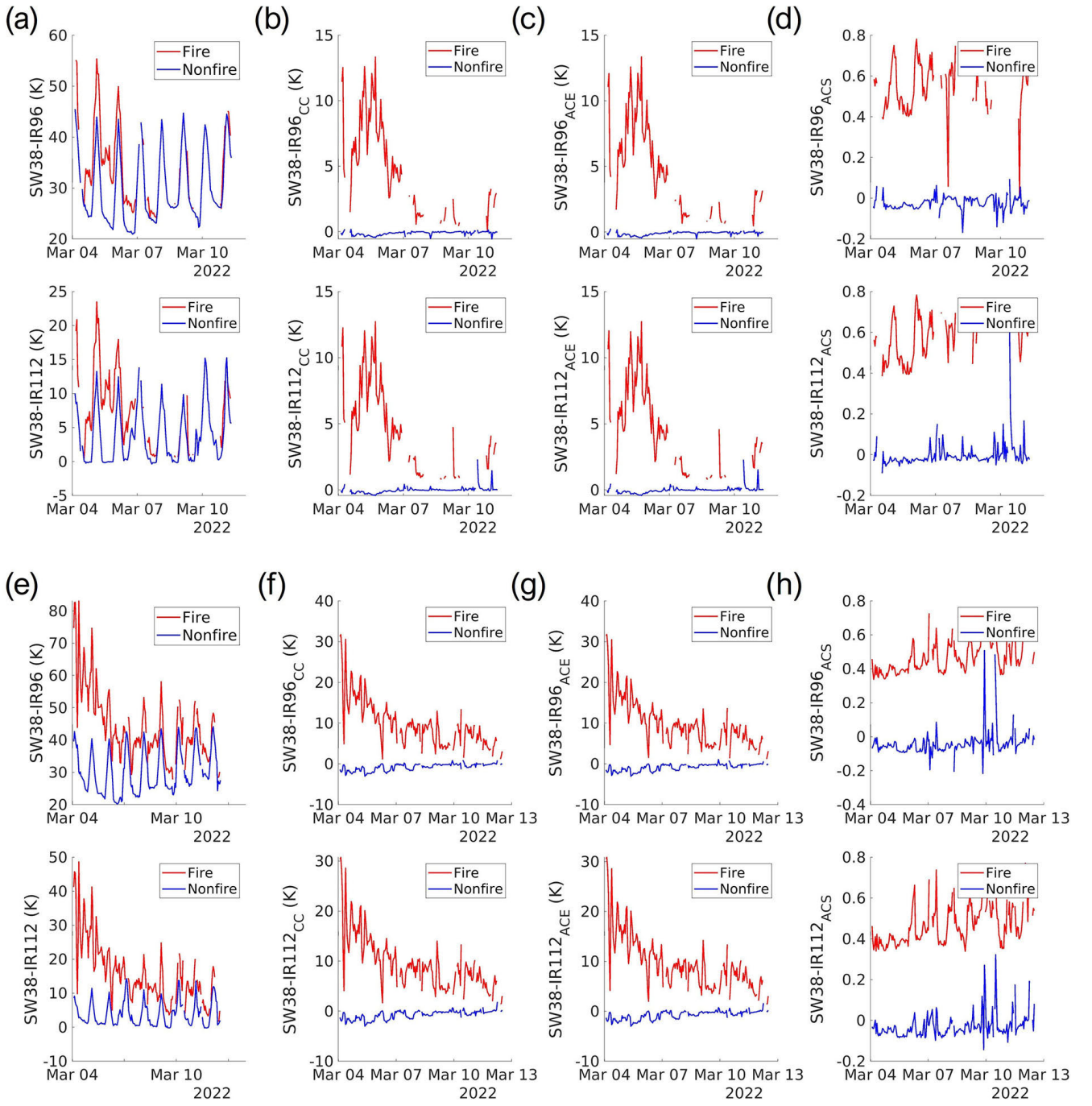


Fig. 12. Time series plot of the input features used in Scheme 3 for (a)–(d) Youngwol and (e)–(h) Uljin cases. ACE and ACS indicate advanced contextual features considering baseline environment and advanced contextual features considering structure similarity, respectively. Note that ACS is relatively independent of wildfire intensity and elapsed time, while ACE is not.

correlation analysis, both DEM and NDVI negatively impact BT. Therefore, if the DEM or NDVI of the target pixel is relatively low compared to its surroundings, the value of the contextual feature will increase, leading to FP even in the absence of wildfires. In contrast, if the DEM or NDVI is relatively high, the value of the contextual feature decreases, which can obscure the detection of smaller wildfires. These scenarios exemplify fundamental flaws in the assumptions of the contextual algorithm and are recognized as sources of both omission and commission errors in prior research [26], [43], [63].

With the implementation of environmental detrending, the ACE effectively mitigated these issues. The greater the discrepancy between the DEM and NDVI values of the target pixel and its surrounding area, the more pronounced the improvement in confidence. However, the enhancement trend for FN with NDVI detrending was notably weaker [Fig. 11(d)]. This discrepancy arises because nonfire samples were extracted under diverse environmental conditions, while fire samples were derived from specific incidents, leading to a narrower range of environmental scenarios encountered.

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Moreover, although the DEM values remain static regardless of wildfire activity, NDVI values tend to decrease as vegetation burns, diminishing the correlation between contextual differences in NDVI and confidence enhancement.

C. Time Series Analysis

A time series analysis was conducted to evaluate the advantages and disadvantages of each feature across various wildfire intensities and elapsed times (Fig. 12). This analysis focused on two cases: Youngwol (case number 40 in Table S1; see the supplementary material) and Uljin. For this purpose, the hourly average values of the four types of input features (BS, CC, ACE, and ACS) were calculated within a 15×15 window centered on the location of each KFS wildfire.

The BS feature indicated the absolute temperature of a wildfire, which was predictably high for intense wildfires but did not account for the baseline temperature, leading to deviations in the case-specific range and displaying a clear diurnal pattern in the nonfire samples [Fig. 12(a) and (e)]. The CC feature measured the spatial anomaly in wildfire temperature relative to the baseline, offering a more consistent range than BS for nonfire samples by incorporating the baseline temperature [Fig. 12(b) and (f)].

While BS is independent of the surrounding window and effectively detects large-scale wildfires that impact multiple pixels, it has difficulty identifying wildfires with weak signals [19], [35], [64], [65]. On the other hand, CC excels in detecting weak-signal wildfires, but its detection efficiency decreases as the number of pixels affected by the fire increases, which in turn raises the BT value of the surrounding window [18], [65], [66].

ACE retains the spatial anomaly detection characteristics of CC but enhances performance significantly in heterogeneous environmental conditions [Fig. 12(c) and (g)], rendering it an excellent alternative to CC. However, similar to other temperature-based anomaly indices, CC and ACE still present some limitations. ACE exhibited a stable range between -3 and 1 K for nonfire samples, but for fire samples, a pronounced variability in the range was evident depending on the intensity and elapsed time of the fire (i.e., from 0.7 to 13.5 K for Youngwol and from 1.2 to 32.3 K for Uljin). This variability indicates that ACE, although overall effective, displays reduced detection sensitivity for low-intensity wildfires, especially as the wildfire event progresses.

Unlike ACE, the ACS is a structure-based similarity index, not dependent on temperature. Thus, ACS maintains a consistent range (0.3–0.8) across all wildfire samples, irrespective of intensity and elapsed time [Fig. 12(d) and (h)]. However, certain nonfire samples adjacent to fire samples also exhibited ACS values above 0.3, indicating limitations in using ACS alone to accurately differentiate between fire and nonfire samples. When used in conjunction with temperature-based anomaly indices such as BS and ACE, ACS can significantly enhance the generalization performance of wildfire detection models.

VI. Conclusion

In this study, a machine learning-based wildfire detection model was developed, incorporating advanced contextual features that consider both environmental and structural information to surmount the limitations inherent in CC algorithms. The integration of ACE and ACS notably improved the performance across all tested machine learning algorithms, with the LGBM model, which utilized ACE and ACS as input variables, achieving the highest metrics (precision: 98.56%; recall: 97.78%; $F1$-score: 98.17%). By analyzing the spatial distribution of historic wildfires in South Korea and assessing its transferability to SWC, the LGBM model demonstrated superior detection sensitivity compared to the GK-2A FF product.

However, the LGBM model did produce certain FP during daytime hours. Although these were somewhat reduced in Scheme 3, they were not entirely eliminated. The adoption of more sophisticated false alarm rejection strategies, employing machine learning and deep learning techniques, could further diminish these FP [67]. Additionally, discrepancies in nonvegetated area masking, originating from the spatial resolution differences between MODIS LC and GK-2A, led to FN near urban locales. Future use of geostationary satellite images with higher spatial resolutions may aid in mitigating this issue. This study did not account for weather and observation conditions such as relative humidity or satellite zenith angle, which could also affect BT alongside DEM and NDVI [68], [69], [70], [71]; these factors should be considered in future research to enhance the model's generalization performance.

Nevertheless, this study introduced a novel approach by integrating environmental and structural information into wildfire detection, successfully addressing the shortcomings of traditional methods and marking a modest advance in the field of satellite-based wildfire monitoring, shifting from a sole focus on temperature data to a more comprehensive utilization of environmental and structural data.

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